



Dr. Sameer Abdullah
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Takhasusi Street cross Makkeh Road
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Order no.: 63245347
Order received: 02 Aug. 2024
Sample type / Sample collection date:
blood, CentoCard® / 31 Jul. 2024
Report date: 06 Sep. 2024
Report type: Final report



Patient no: **1932432**, First Name: **AMERH ABUBAKUR**, Last Name: **ENAYAT**
DOB: **03 Sep. 2022**, Sex: **female**, Your ref.: **20030712**

Additional report recipient(s): Dr. Fouad Alghamdi, Al Salam Hospital, Neuropediatrics, KHOBAR, Saudi Arabia

Test(s) requested: CentoXome® MOx 1.0 Trio

CLINICAL INFORMATION

Abnormal cerebral ventricle morphology; Acidemia; Frontal bossing; Macrocephaly; Methylmalonic acidemia; Ventriculomegaly
(Clinical information indicated above follows HPO nomenclature.)
Family history: Unknown.
Consanguineous parents: Unknown.



POSITIVE RESULT
Likely pathogenic variant identified

INTERPRETATION

A heterozygous likely pathogenic variant was identified in the *PTCH1* gene. **The result is consistent with a genetic diagnosis of autosomal dominant *PTCH1*-related disease.** The familial segregation analysis suggests the identified variant is *de novo* (although, other less likely explanations are possible, e.g., germline mosaicism in one of the parents).

RECOMMENDATIONS

- Targeted testing for all affected and at-risk family members, if any, is recommended.
- Genetic counselling, including reproductive counselling (discussing prenatal and preimplantation diagnoses, if relevant), is recommended.

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MAIN FINDINGS

SEQUENCE VARIANTS							
GENE	VARIANT COORDINATES	AMINO ACID CHANGE	SNP IDENTIFIER	ZYGOSITY	IN SILICO PARAMETERS*	ALLELE FREQUENCIES**	TYPE AND CLASSIFICATION***
PTCH1	NM_000264.5:c.92_98del	p.(Gly31Alafs*47)	N/A	Heterozygous	PolyPhen: N/A Align-GVDG: N/A SIFT: N/A MutationTaster: N/A Conservation_nt: Conservation_aa:	gnomAD: - ESP: - 1000 G: - CentoMD: -	Frameshift Likely pathogenic (class 2)

Variant annotation based on CentoCloud Bioinformatics pipeline. * AlignGVD: C0: least likely to interfere with function, C65: most likely to interfere with function; splicing predictions: Ada and RF scores. ** Genome Aggregation Database (gnomAD), Exome Sequencing Project (ESP), 1000Genome project (1000G) and CentoMD® (latest database available). *** based on ACMG recommendations.

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VARIANT INTERPRETATION

PTCH1, c.92_98del p.(Gly31Alafs*47)

The PTCH1 variant c.92_98del p.(Gly31Alafs*47) creates a shift in the reading frame starting at codon 31 in exon(s) no. 1 (of 24). To the best of our knowledge this is a novel variant not previously reported in the literature. It is classified as likely pathogenic based on CENTOGENE's implementation of the ACMG/AMP/ClinGen SVI guidelines.

The basal cell nevus syndrome 1, also referred as Gorlin Syndrome (GS), is characterized by numerous basal cell cancers and epidermal cysts of the skin, calcified dural folds, keratocysts of the jaws, palmar and plantar pits, ovarian fibromas, medulloblastomas, lymphomesentreic cysts, fetal rhabdomyomas, and various stigmata of maldevelopment (e.g., rib and vertebral abnormalities, cleft lip or cleft palate, and cortical defects of bones) (PMID:11932998). GS is characterized by the early onset of mandibular odontogenic keratocysts (2nd decade of life) and/or multiple BCCs (most commonly seen on the face, back and chest - 3rd decade of life). Approximately 60% of individuals have a recognizable appearance with macrocephaly, frontal bossing, coarse facial features, and facial milia. Ectopic calcification, particularly in the falx, is present in more than 90% of affected individuals by age 20 years. Palmar or plantar pits (asymmetrical, 2-3 mm in diameter, 1-3 mm in depth and developing in the 2nd decade) and skeletal anomalies (fusion of vertebrae, wedge-shaped vertebrae, bifid or fused ribs, hemivertebra, kyphoscoliosis, pectus deformity, Sprengel deformity, syndactyly, polydactylia) are also reported. Additional features include facial dysmorphism (cleft lip/palate, macrocephaly), eye anomalies (cataract, coloboma, microphthalmos) and lymphomesenteric cysts. A predisposition to malignant or benign tumors such as medulloblastoma, meningioma, papillary fibroelastoma of the heart, ovarian fibroma (commonly bilateral and calcified), fibrosarcoma, nephroblastoma is observed (ORPHA:377). Mode of Inheritance: Autosomal dominant. (OMIM®: 109400).

Holoprosencephaly (HPE) is the most commonly occurring congenital structural forebrain anomaly in humans. HPE is associated with mental retardation and craniofacial malformations. Considerable heterogeneity in the genetic causes of HPE has been demonstrated (Ming et al., 2002; PMID:11941477). Mode of Inheritance: Autosomal dominant (OMIM®: 610828)

SECONDARY FINDINGS

If consent is provided, in line with ACMG recommendations (ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing; Genetics in Medicine, 2023; PMID: 37347242) we report secondary findings, i.e. relevant pathogenic and likely pathogenic variants in the recommended genes for the indicated phenotypes in this publication.

We did not detect any relevant variants in the genes for which secondary findings are reported.

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CARRIERSHIP FINDINGS

In this table we list sequence variants previously ascertained or evaluated and classified in CENTOGENE as "pathogenic" and "likely pathogenic", in selected genes associated with recessive severe and early-onset Mendelian diseases. As only in-house classified variants are presented, it should not be considered a comprehensive list of variants in these genes and does not provide a complete list of potentially relevant genetic variants in the patient. The complete gene list can be found at www.centogene.com/carriership-findings (please contact CENTOGENE customer support if the gene list has been updated after this report was issued). Orthogonal validation was not performed for these variants. Therefore, if any variant is used for clinical management of the patient, confirmation by another method needs to be considered. Furthermore, the classification of these variants may change over time, however reclassification reports for these variants will not be issued. CENTOGENE is not liable for any missing variant in this list and/or any provided classification of the variants at a certain point of time. As the identified variants may indicate (additional) genetic risks or diagnoses in the patient and/or family and/or inform about reproductive risks, we recommend discussing these findings in the context of genetic counselling.

In accordance with this disclaimer no relevant pathogenic or likely pathogenic variant was identified.

CENTOGENE VARIANT CLASSIFICATION (BASED ON ACMG RECOMMENDATIONS)

Class 1 – Pathogenic

Class 2 – Likely pathogenic

Class 3 – Variant of uncertain significance (VUS)

Class 4 – Likely benign

Class 5 – Benign

Additionally, other types of clinically relevant variants can be identified (e.g. risk factors, modifiers).

METHODS

CentoXome® MOx 1.0 Trio

Genomic DNA is enzymatically fragmented, and target regions are enriched using DNA capture probes. These regions include approximately 41 Mb of the human coding exome (targeting > 98% of the coding RefSeq from the human genome build GRCh37/hg19), as well as the mitochondrial genome. The generated library is sequenced on an Illumina platform to obtain at least 20x coverage depth for > 98% of the targeted bases. An in-house bioinformatics pipeline, including read alignment to GRCh37/hg19 genome assembly and revised Cambridge Reference Sequence (rCRS) of the Human Mitochondrial DNA (NC_012920), variant calling, annotation, and comprehensive variant filtering is applied. All variants with minor allele frequency (MAF) of less than 1% in gnomAD database, and disease-causing variants reported in HGMD®, in ClinVar or in CENTOGENE's in-house Biodatabank are evaluated. The investigation for relevant variants is focused on coding exons and flanking +/-10 intronic nucleotides of genes with clear gene-phenotype evidence (based on OMIM® information). All potential patterns for mode of inheritance are considered. In addition, provided family history and clinical information are used to evaluate identified variants with respect to their pathogenicity and disease causality. Variants are categorized into five classes (pathogenic, likely pathogenic, variant of uncertain significance [VUS], likely benign, and benign) according to ACMG/AMP guidelines for classification of variants in addition to ClinGen recommendations. All relevant variants related to the phenotype of the patient are reported. For CentoXome® MOx, if applicable, biochemical analysis is performed upon detection of relevant variants by sequencing. This enhances the diagnosis of metabolic disorders, optimizes variant classification, and helps to ascertain the eventual contribution to the phenotype; the list of enzyme-activity assays and biomarkers can be obtained at www.centogene.com/mox. CENTOGENE has established stringent quality criteria and validation processes for variants detected by NGS. Variants with low sequencing quality and/or unclear zygosity are confirmed by orthogonal methods. Consequently, a specificity of > 99.9% for all reported variants is warranted. Mitochondrial variants are reported for heteroplasmy levels of 15% or higher. The copy number variation (CNV) detection software has a sensitivity of more than 95%. For the uniparental disomy (UPD) screening, a specific algorithm is used to assess the well-known clinically relevant chromosomal regions (6q24, 7, 11p15.5, 14q32, 15q11q13, 20q13 and 20).

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ANALYSIS STATISTICS

CentoXome® MOx 1.0 Trio

Targeted nucleotides covered	≥ 20x	99.49%
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LIMITATIONS

CentoXome® MOx 1.0 Trio

The genetic results are interpreted in the context of the provided clinical findings, family history, and other laboratory data. Only variants in genes potentially related to the proband's medical condition are reported. Misinterpretation of results may occur if the provided genetic data or patient information is inaccurate and/or incomplete. If the obtained genetic results are not compatible with the clinical findings, additional testing should be considered.

The genes with mapping issues in GRCh37/hg19 genome assembly, the non-protein-coding disease-associated genes, and approximately 0.2 Mb of genomic regions that are hard to sequence by current enrichment technology and are without evidenced relevance for monogenic disorders, are excluded from this analysis. More complex genetic events such as inversions, translocations, and repeat expansions, are not analyzed in this test. The UPD detection is a screening method, and therefore false-positive and false-negative results may occur. In addition, due to technology limitations, certain regions may be poorly covered, or not covered at all. In these regions and others encompassing repetitive, high-homology (such as pseudogene homology), and GC-rich sequences, relevant variants can be missed. Extremely low-coverage calls (homo/hemizygous or heterozygous calls with less than three or four reads, respectively) are expected to be artifacts based on our extensive validations and are consequently not considered during the analysis. The CNV detection sensitivity is decreased for repetitive and homologous regions such as pseudogenes, as well as for events spanning two or less exons. Mitochondrial variants with heteroplasmy levels below 15% may not be detected. It is expected that lower quality samples (prenatal, product of conception, blood from patients with hematologic disorders, and highly degraded DNA) may generate lower quality NGS data; in these cases, CNV analysis and/or mitochondrial genome analysis may not be possible to perform. Potential aberrant splicing is assessed with splice prediction tools. Intronic variants that are beyond 10 nucleotides from exon-intron boundaries are not considered for aberrant splicing analysis, with the exception of known pathogenic splicing variants evidenced by external sources.

ADDITIONAL INFORMATION

This test was developed, and its performance was validated, by CENTOGENE. The US Food and Drug Administration (FDA) has determined that clearance or approval of this method is not necessary and thus neither have been obtained. This test has been developed for clinical purposes. All test results are reviewed, interpreted and reported by our scientific and medical experts.

To exclude mistaken identity in your clinic, several guidelines recommend testing a second sample that is independently obtained from the proband. Please note that any further analysis will result in additional costs.

The classification of variants can change over the time. Please feel free to contact CENTOGENE (customer.support@centogene.com) in the future to determine if there have been any changes in classification of any reported variants.

DISCLAIMER

Any preparation and processing of a sample from patient material provided to CENTOGENE by a physician, clinical institute or a laboratory (by a "Partner") and the requested genetic and/or biochemical testing itself is based on the highest and most current scientific and analytical standards. However, in very few cases genetic or biochemical tests may not show the correct result, e.g. because of the quality of the material provided by a Partner to CENTOGENE or in cases where any test provided by CENTOGENE fails for unforeseeable or unknown reasons that cannot be influenced by CENTOGENE in advance. In such cases, CENTOGENE shall not be responsible and/or liable for the incomplete, potentially misleading or even wrong result of any testing if such issue could not be recognized by CENTOGENE in advance.

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